Practice workbook · Explain how tissue fluid forms, what it does and why it builds up

Pressure at the capillary bed pushing fluid out and drawing it back, its composition compared with blood, exchange with cells, lymph drainage and swelling.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-TISSUE-FLUID-FORMATION-AND-FUNCTION

1. Question 8 · part b · 4 marks

November 2024 Paper 1

Worked-example retry · Attempt part b only.

8 Body cells are surrounded by tissue fluid.

(a) Describe the function of tissue fluid.

(2)

(b) Explain how tissue fluid is formed.

(4)

(c) Some people have a condition called cellular oedema.

In these people too much water enters body cells.

Explain how water can enter body cells from the surrounding tissue fluid.

(3)

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1. Question 8 · part b · 4 marks · Working and answer

Working and answer

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2. Question 4 · part a.ii · 4 marks

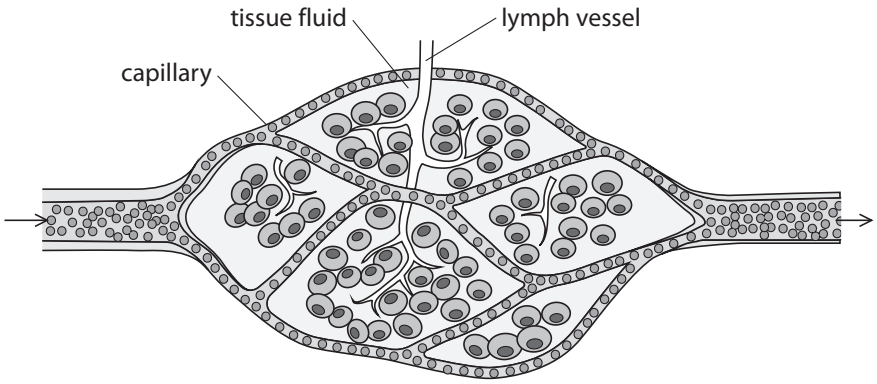
January 2023 Paper 2

Further practice · Attempt part a.ii only.

4 (a) When blood flows into a network of capillaries, tissue fluid is produced.

The tissue fluid supplies the surrounding cells with molecules.

The diagram shows a network of capillaries and surrounding cells.



(i) Give two differences between the composition of tissue fluid and the composition of blood entering the capillary network.

(2)

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(ii) Describe how tissue fluid is formed.

(4)

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2. Question 4 · part a.ii · 4 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

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(iii) Lymphoedema occurs when tissue fluid builds up causing swelling.

This is more common in the legs and arms.

One treatment for this is to increase physical activity, resulting in more muscle contraction.

Explain why increased muscle contraction could reduce the swelling caused by the build up of tissue fluid.

(2)

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(b) Lymphocytes, found in lymph nodes, produce antibodies in response to specific antigens.

Explain why the reaction between an antigen and an antibody is specific.

(3)

(Total for Question 4 = 11 marks)

2. Question 4 · part a.ii · 4 marks · Working and answer

Working and answer

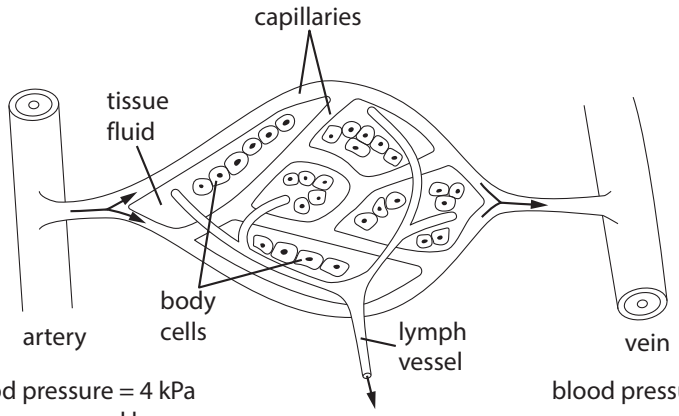
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3. Question 7 · part a.i · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.i only.

7 The diagram shows a capillary network.



(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

(ii) Explain why the pressure caused by the plasma proteins does not change.

(2)

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3. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

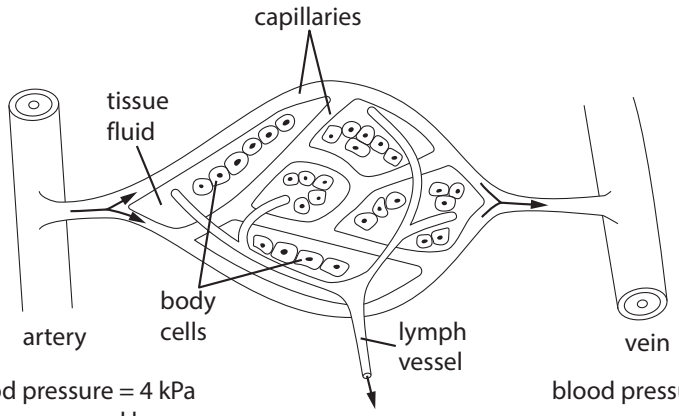
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4. Question 7 · part a.ii · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

7 The diagram shows a capillary network.



blood pressure = 4 kPa
pressure caused by plasma proteins = 3.3 kPa

blood pressure = 1.6 kPa
pressure caused by plasma proteins = 3.3 kPa

(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

(ii) Explain why the pressure caused by the plasma proteins does not change.

(2)

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4. Question 7 · part a.ii · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

(b) Describe how a sample of plasma could be tested for the presence of proteins.

(4)

(c) Tissue fluid is formed by the movement of molecules from the blood. This occurs in the capillary network.

Use data from the diagram to determine the direction of movement of fluid between the capillaries and tissue fluid at the artery end and at the vein end of the capillary network.

(4)

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4. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

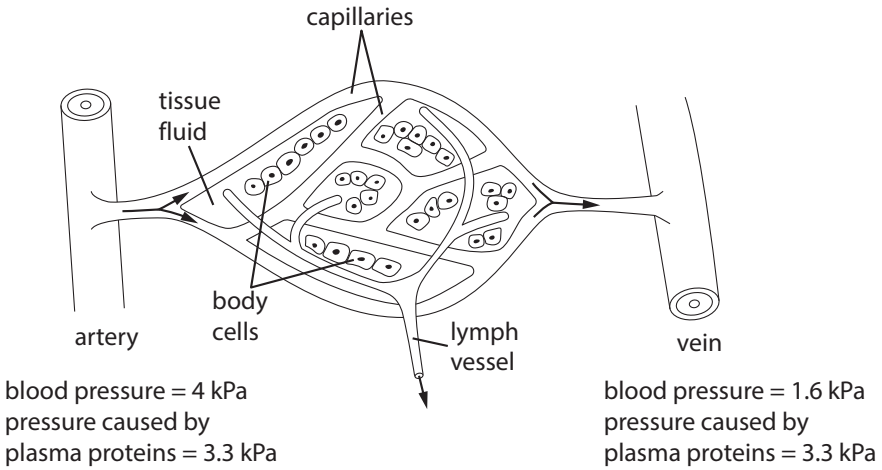
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5. Question 7 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

7 The diagram shows a capillary network.



(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

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(2)

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5. Question 7 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

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(b) Describe how a sample of plasma could be tested for the presence of proteins.

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(c) Tissue fluid is formed by the movement of molecules from the blood. This occurs in the capillary network.

Use data from the diagram to determine the direction of movement of fluid between the capillaries and tissue fluid at the artery end and at the vein end of the capillary network.

(4)

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5. Question 7 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

(d) A condition known as leaky capillary syndrome occurs in some people. When this occurs, much more of the fluid from the capillary passes out into the tissue fluid.

Suggest what effect leaky capillary syndrome would have on the circulatory system.

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(Total for Question 7 = 14 marks)

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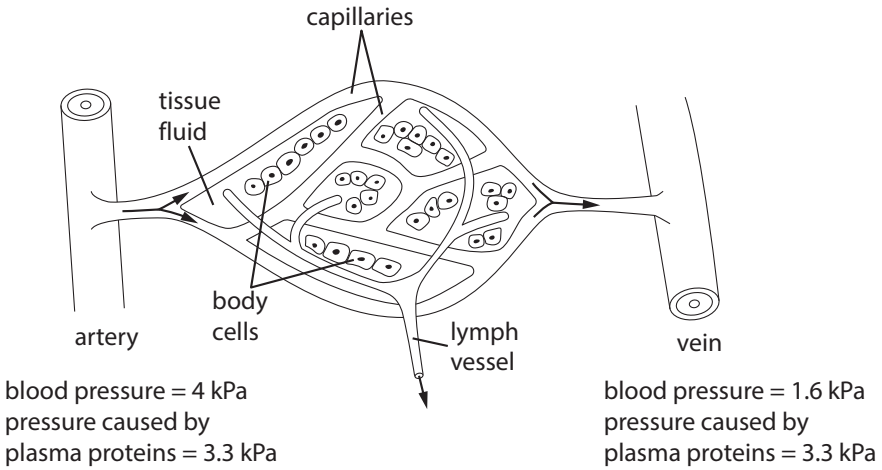
Working and answer

6. Question 7 · part d · 2 marks

June 2025 Paper 1

Further practice · Attempt part d only.

7 The diagram shows a capillary network.



(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

(ii) Explain why the pressure caused by the plasma proteins does not change.

(2)

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6. Question 7 · part d · 2 marks

June 2025 Paper 1

Further practice · Attempt part d only.

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(b) Describe how a sample of plasma could be tested for the presence of proteins.

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(c) Tissue fluid is formed by the movement of molecules from the blood. This occurs in the capillary network.

Use data from the diagram to determine the direction of movement of fluid between the capillaries and tissue fluid at the artery end and at the vein end of the capillary network.

(4)

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6. Question 7 · part d · 2 marks

June 2025 Paper 1

Further practice · Attempt part d only.

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Suggest what effect leaky capillary syndrome would have on the circulatory system.

(2)

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(Total for Question 7 = 14 marks)

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Working and answer

7. Question 4 · part a.iii · 2 marks

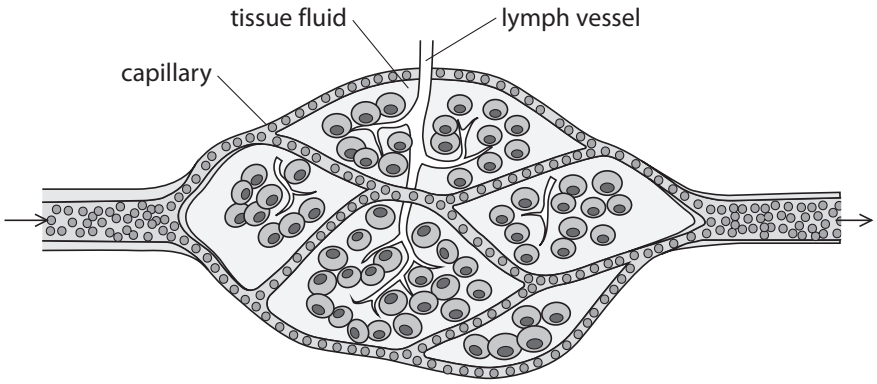
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Further practice · Attempt part a.iii only.

4 (a) When blood flows into a network of capillaries, tissue fluid is produced.

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7. Question 4 · part a.iii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.iii only.

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(iii) Lymphoedema occurs when tissue fluid builds up causing swelling.

This is more common in the legs and arms.

One treatment for this is to increase physical activity, resulting in more muscle contraction.

Explain why increased muscle contraction could reduce the swelling caused by the build up of tissue fluid.

(2)

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(b) Lymphocytes, found in lymph nodes, produce antibodies in response to specific antigens.

Explain why the reaction between an antigen and an antibody is specific.

(3)

(Total for Question 4 = 11 marks)

7. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

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8. Question 4 · part a.i · 2 marks

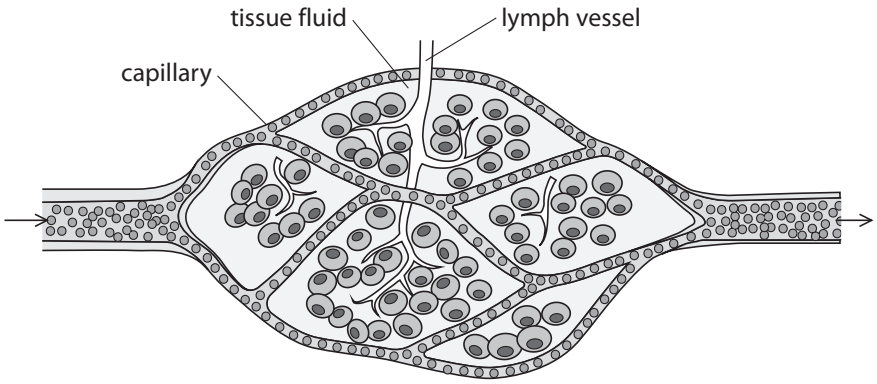
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Further practice · Attempt part a.i only.

4 (a) When blood flows into a network of capillaries, tissue fluid is produced.

The tissue fluid supplies the surrounding cells with molecules.

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8. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

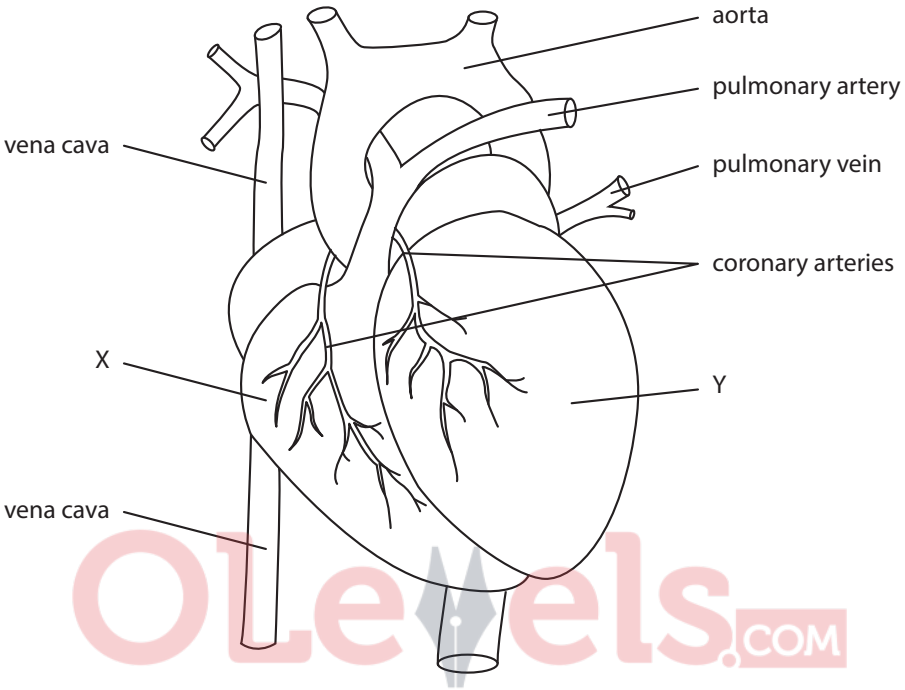
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9. Question 7 · part d · 3 marks

November 2023 Paper 1

Further practice · Attempt part d only.

7 The diagram shows the heart as seen from the front.



(a) The vena cava carries deoxygenated blood.

Name a different blood vessel, shown on the diagram, that also carries deoxygenated blood.

(1)

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9. Question 7 · part d · 3 marks

November 2023 Paper 1

Further practice · Attempt part d only.

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(b) Blood chambers X and Y have walls that are different in thickness.

Explain this difference.

(5)

(c) The coronary arteries can become blocked.

Explain the effect of this blockage on the functioning of the heart.

(3)

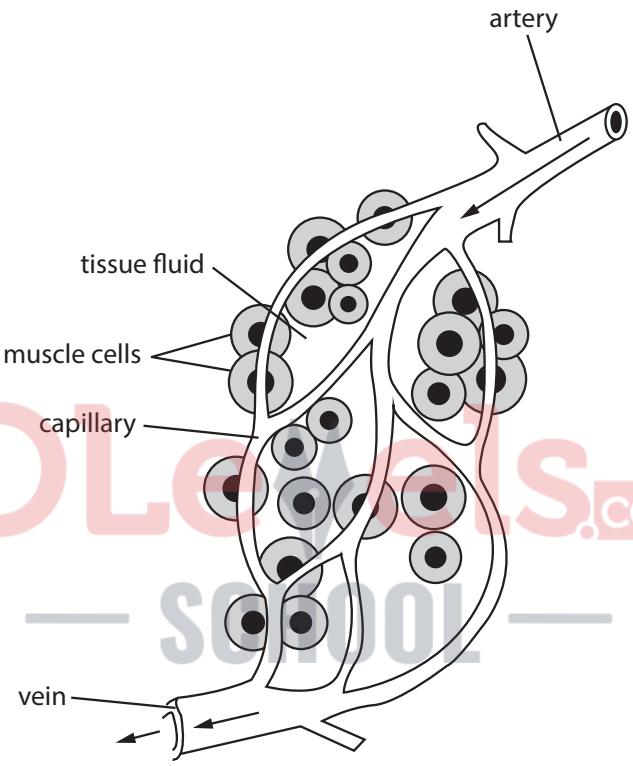
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9. Question 7 · part d · 3 marks

November 2023 Paper 1

Further practice · Attempt part d only.

(d) The diagram shows a capillary bed.



Use the diagram to explain the function of the tissue fluid.

(3)

(Total for Question 7 = 12 marks)

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9. Question 7 · part d · 3 marks · Working and answer

Working and answer

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10. Question 8 · part a · 2 marks

November 2024 Paper 1

Further practice · Attempt part a only.

8 Body cells are surrounded by tissue fluid.

(a) Describe the function of tissue fluid.

(2)

(b) Explain how tissue fluid is formed.

(4)

(c) Some people have a condition called cellular oedema.

In these people too much water enters body cells.

Explain how water can enter body cells from the surrounding tissue fluid.

(3)

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10. Question 8 · part a · 2 marks · Working and answer

Working and answer

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